SecureCorp Cybersecurity Lab

Technical Report

Project: SecureCorp Cybersecurity Lab

Domain: Cybersecurity — Active Directory — SIEM — Endpoint Monitoring — Offensive Security

Program: Cybersecurity and Embedded Systems Engineering

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Executive Summary

The SecureCorp Cybersecurity Lab is a virtualized environment that reproduces a simplified enterprise infrastructure in order to study security monitoring from the perspective of an analyst. It brings together an OPNsense firewall, an Active Directory domain, a Windows 11 endpoint, a Kali Linux machine and a Wazuh monitoring platform.

The solution implements the following security mechanisms:

- Network segmentation
- Centralized identity management
- Active Directory analysis
- Endpoint monitoring
- Collection of Windows events
- Sysmon monitoring
- File Integrity Monitoring (FIM)
- Security Configuration Assessment (SCA)
- System inventory
- Controlled network reconnaissance
- Authentication monitoring

The scenarios carried out demonstrate the lab's ability to generate and centrally collect security events. The project therefore serves as a practical learning and demonstration platform for Blue Team operations, SOC and SIEM work, Active Directory security, and controlled offensive security.

1. Introduction

The growing number of cyberattacks against modern IT infrastructures makes it essential to put in place mechanisms that prevent, monitor and analyze suspicious activity.

In this context, the **SecureCorp Cybersecurity Lab** project consists of designing a virtualized cybersecurity lab that reproduces a simplified enterprise infrastructure.

The lab combines several security technologies:

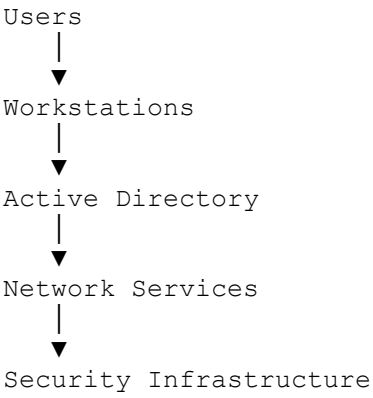
- OPNsense for network routing and filtering
- Active Directory for centralized identity management
- Windows 11 as the client workstation
- Kali Linux as the offensive analysis machine
- Wazuh as the monitoring and detection platform
- Sysmon to enrich Windows events
- File Integrity Monitoring (FIM) to monitor file integrity
- Security Configuration Assessment (SCA) to assess system configuration
- Syscollector for system inventory
- BloodHound for Active Directory structure analysis
- Nmap for network reconnaissance

The main objective is to build an infrastructure in which activity carried out on the network can be observed from the perspective of a security analyst.

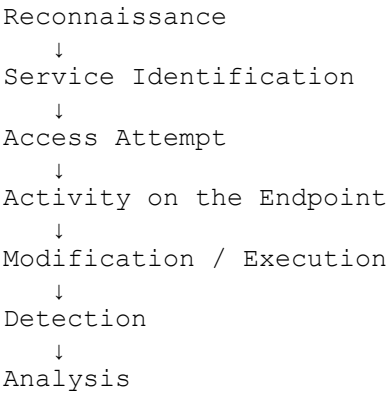
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2. Context and Problem Statement

An enterprise infrastructure generally consists of several layers:



A compromise, however, typically unfolds in several stages:



The problem addressed by this project is therefore the following:

How can a virtualized enterprise environment be built to generate, collect and analyze security events originating from both the network and the endpoints?

SecureCorp provides a practical answer to this question.

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3. Project Objectives

3.1 General Objective

Design an enterprise cybersecurity lab that reproduces controlled scenarios of administration, reconnaissance and security monitoring.

3.2 Specific Objectives

The specific objectives are to:

- Build a segmented network infrastructure

- Configure a firewall and router with OPNsense
- Deploy an Active Directory domain
- Create users, groups and organizational units
- Analyze Active Directory with BloodHound
- Install Wazuh
- Connect a Windows endpoint to Wazuh
- Install Sysmon
- Collect Windows events
- Monitor file integrity
- Assess the security configuration
- Inventory installed software and system components
- Perform network reconnaissance from Kali
- Generate a controlled authentication attempt
- Observe the events produced
- Document selected activities according to MITRE ATT&CK

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4. General Overview of the Lab

The infrastructure is composed of the following main components:

Component	Role
Kali Linux	Analysis and reconnaissance machine
OPNsense	Firewall and router
SERVER01	Active Directory domain controller
WCLIENT	Windows client workstation
Wazuh	Monitoring and detection
BloodHound	Active Directory analysis
VirtualBox	Hypervisor



Figure 1 — VirtualBox console showing the five lab virtual machines in the running state

The lab is isolated within a virtualized environment so that security tests can be carried out without affecting any real infrastructure.

5. Overall Architecture

The logical architecture of the lab is represented below.

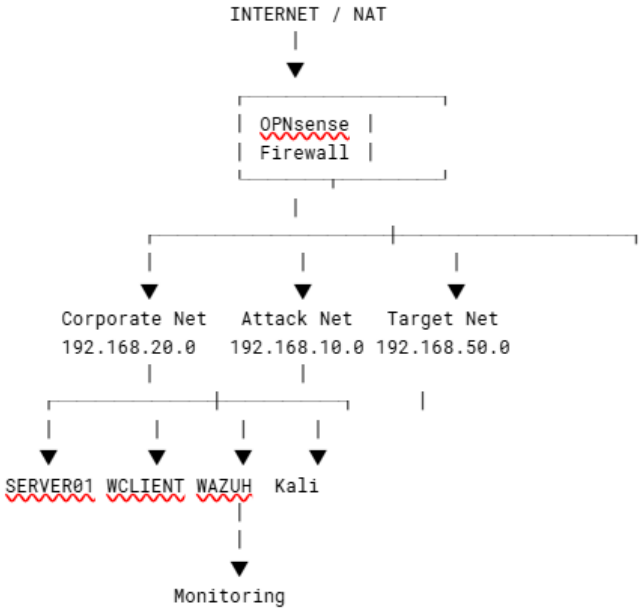


Figure 2 — Logical architecture of the SecureCorp lab

6. Network Architecture

The lab uses several virtual networks to separate the different roles.

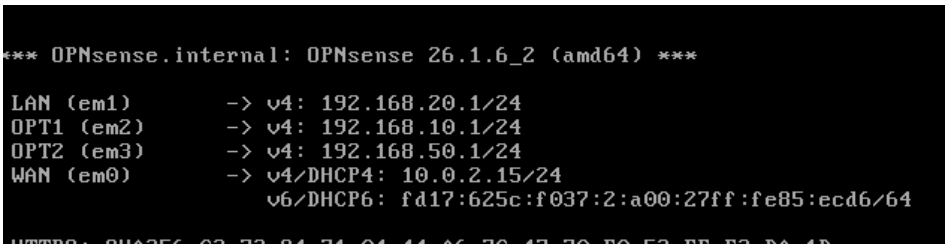


Figure 3 — OPNsense console showing the interface and address assignments

Interface	Device	Address	Role
LAN	em1	192.168.20.1/24	Corporate network
OPT1	em2	192.168.10.1/24	Attack network
OPT2	em3	192.168.50.1/24	Target network
WAN	em0	10.0.2.15/24 (DHCP)	Internet / NAT

7. Virtual Machines

The table below lists the machines deployed in the lab.

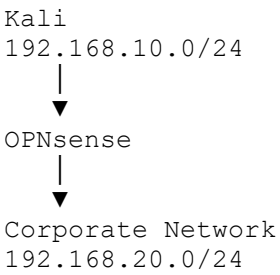
Machine	System	IP Address	Function
Kali	Kali Linux	192.168.10.100	Reconnaissance
WCLIENT	Windows 11	192.168.20.20	Endpoint
WAZUH	Ubuntu Server	192.168.20.116	SIEM / Monitoring
SERVER01	Windows Server	192.168.20.10	Active Directory

8. OPNsense Configuration

OPNsense is the central network component of the lab. Its main functions are:

- Inter-network routing
- Filtering
- Traffic control
- Segmentation
- Interface management
- Observation of communications

The architecture relies on the following principle:



This separation reproduces a situation in which the analysis machine is located on a different network from the target machine.

9. Active Directory

Active Directory provides the lab's centralized identity management.

The domain is `securecorp.local` and the domain controller is `SERVER01`.

Active Directory notably allows the management of:

- Users
- Groups
- Computers

- Organizational units
- Policies
- Relationships between objects

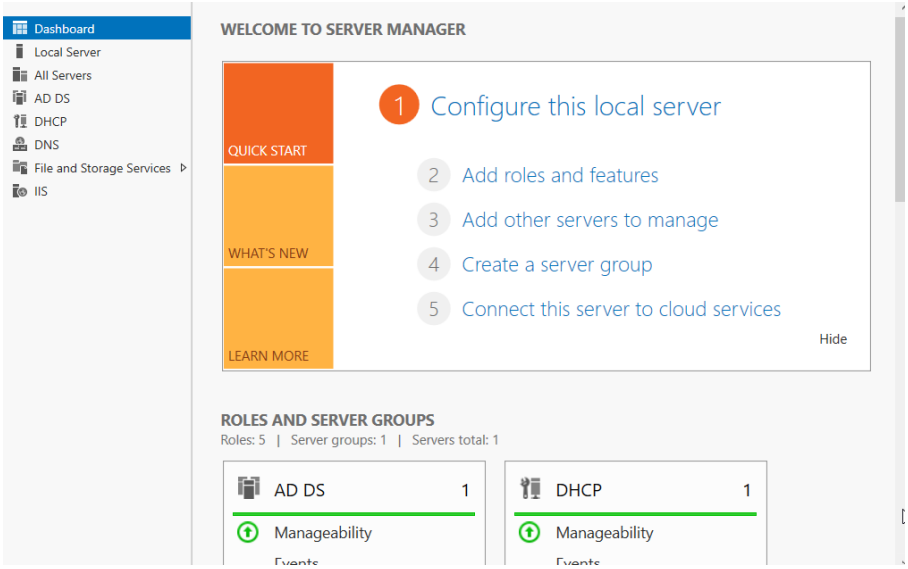


Figure 4 — Server Manager on SERVER01 showing the AD DS and DHCP roles

---

10. Domain Structure

The logical structure of the domain is represented as follows:

```
securecorp.local
│
├── Admins
├── Builtin
├── Computers
├── Domain Controllers
├── ForeignSecurityPrincipals
├── Groups
├── Managed Service Accounts
├── SecureCorp_Users
├── Servers
├── service_Accounts
├── Users
└── Workstations
```

Creating an organizational structure makes it possible to separate resources more effectively and simplifies policy application.

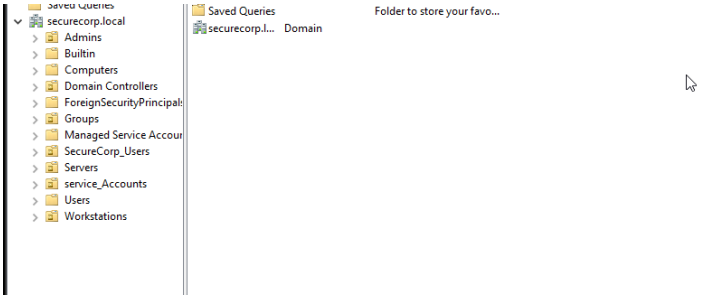


Figure 5 — Active Directory Users and Computers — domain structure of securecorp.local

11. Users and Groups

User accounts and groups were configured to reproduce the organization of a company. Separating groups makes it possible to manage permissions centrally.

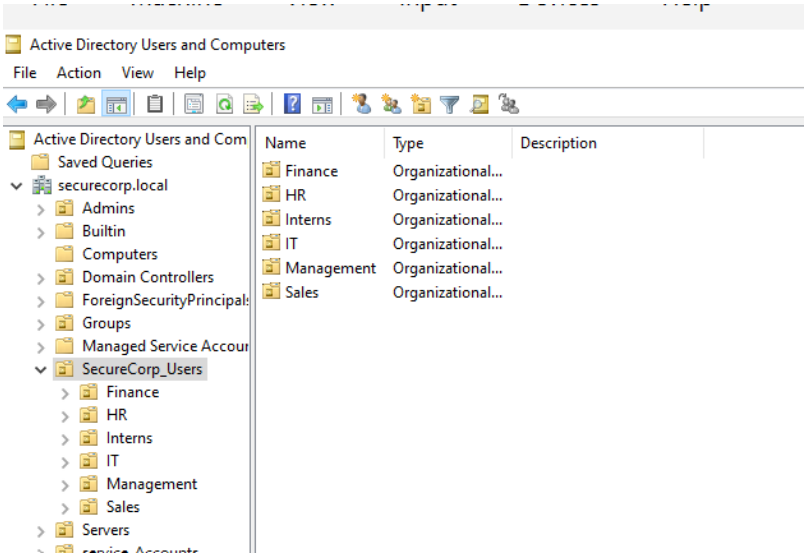


Figure 6 — Organizational units under SecureCorp_Users

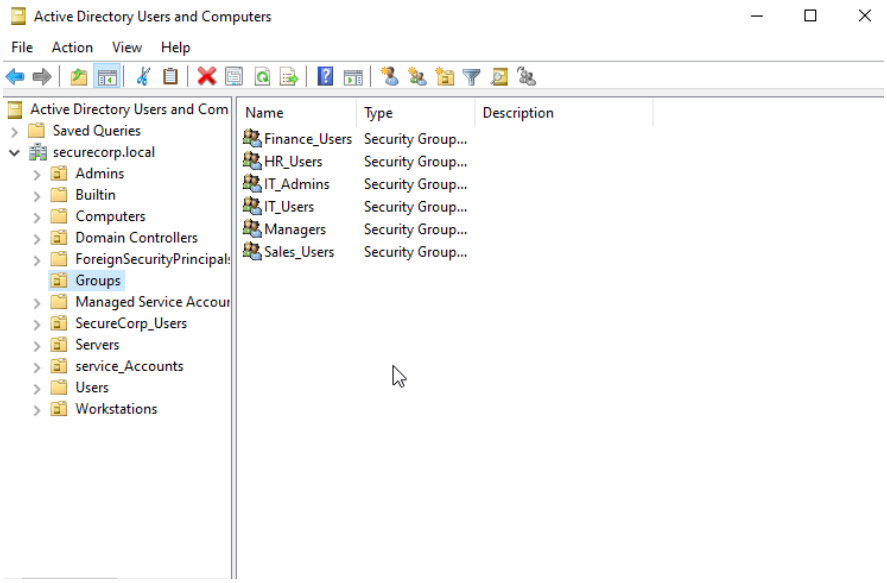


Figure 7 — Security groups defined in the Groups container

12. Organizational Units

Organizational Units (OUs) are used to structure Active Directory objects. They make it possible to:

- Organize users
- Organize computers
- Apply Group Policy Objects (GPOs)
- Delegate administration

In the lab, the SecureCorp_Users OU contains one sub-OU per department:

```
securecorp.local
└── SecureCorp_Users
    ├── Finance
    ├── HR
    ├── Interns
    ├── IT
    ├── Management
    └── Sales
```

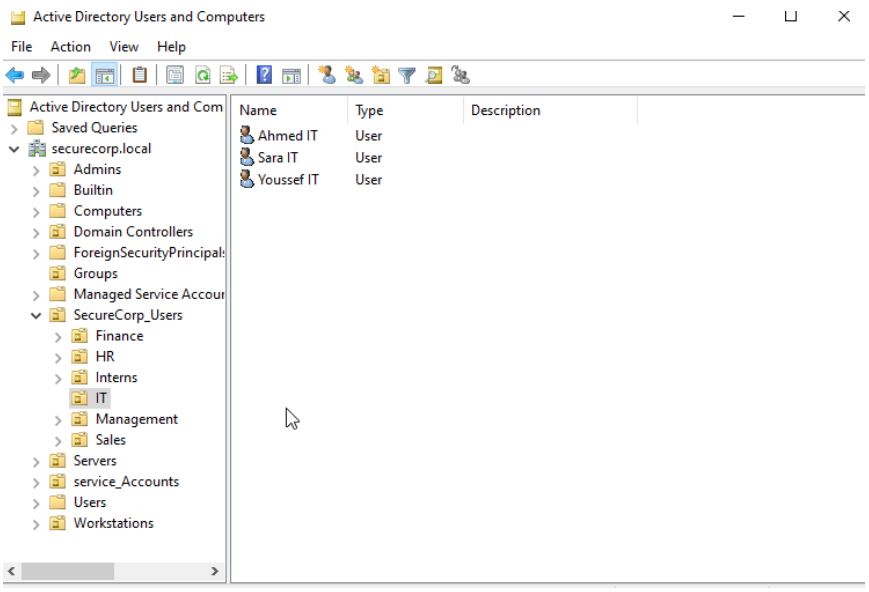


Figure 8 — User accounts contained in the IT organizational unit

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13. Group Policy

Group Policy Objects (GPOs) make it possible to centralize security configuration. In the lab, they are the mechanism used to reproduce centralized administration of Windows workstations.

They can notably control:

- Password policies
- Security settings
- Logging
- Restrictions
- Windows configuration

---

14. BloodHound

BloodHound was used to analyze the structure of the Active Directory environment. Its purpose is to visualize, in the form of a graph, the relationships between:

- Users
- Groups
- Computers
- OUs

- GPOs
- Domains

Data collection was performed from Kali with bloodhound-python and produced the following categories of data: computers, containers, domains, GPOs, groups, OUs and users.

Object type	Objects collected
Domains	1
Computers	2
Users	15
Groups	58
GPOs	2
OUs	13
Containers	19
Trusts	0

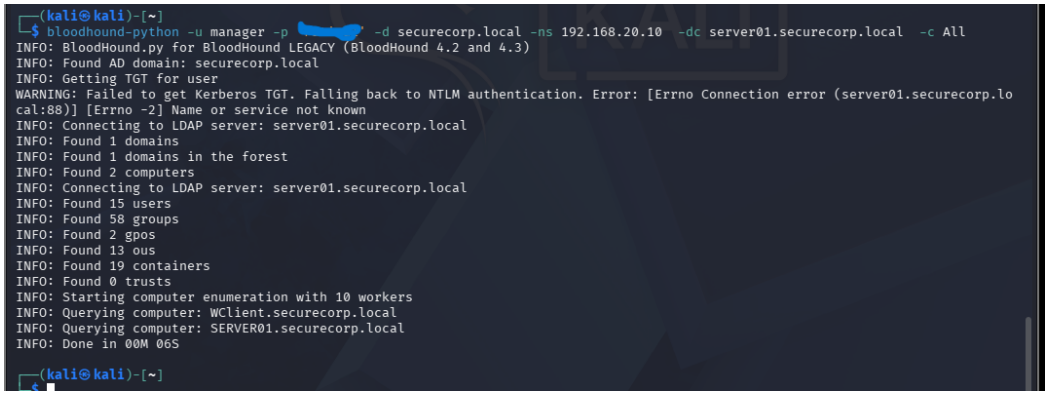


Figure 9 — BloodHound collection performed from Kali with bloodhound-python

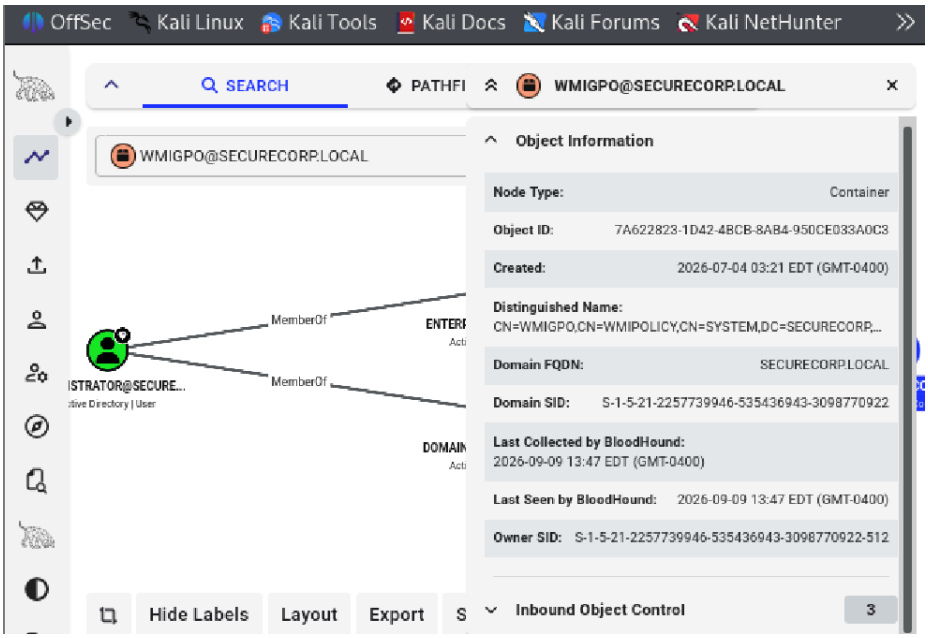


Figure 10 — BloodHound interface — object information for the WMIGPO container

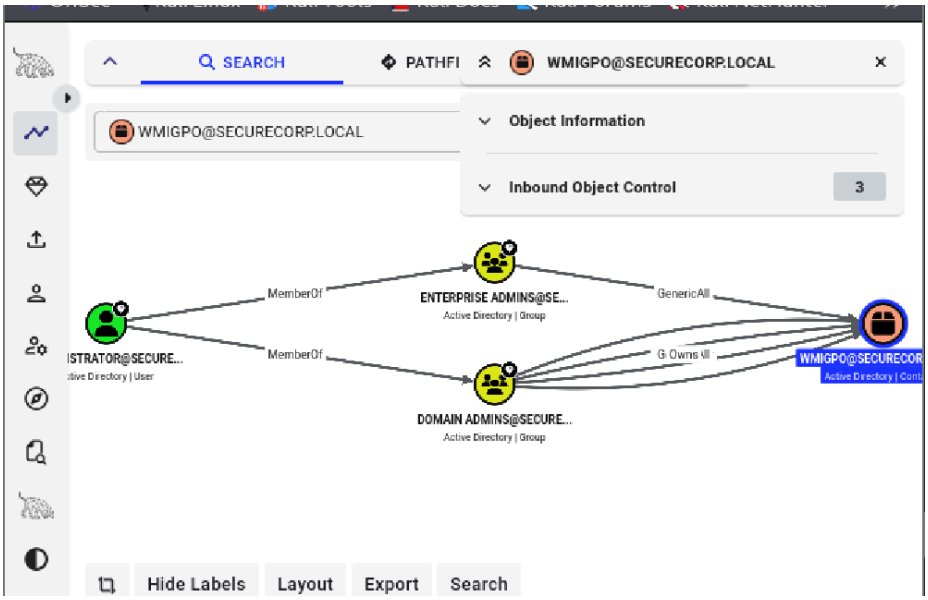


Figure 11 — Graph of relationships towards the WMIGPO container

BloodHound analysis therefore provides a graphical representation of the Active Directory environment.

---

15. Wazuh Deployment

Wazuh is the central monitoring component of the lab. It collects and analyzes data coming from the endpoints.

The architecture is as follows:

```
WCLIENT
   |  Wazuh Agent
   ▼
Wazuh Manager
   |
   ▼
Wazuh Indexer
   |
   ▼
Wazuh Dashboard
```

The Wazuh server uses the address `192.168.20.116`.

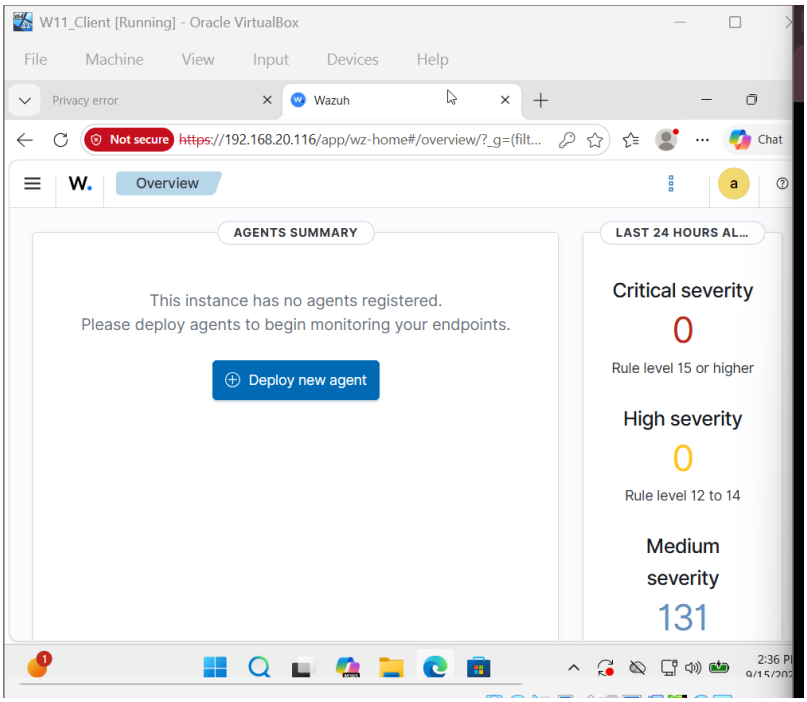


Figure 12 — Wazuh dashboard accessed from WCLIENT, before agent enrollment

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16. Wazuh Agent Deployment

The Wazuh agent was installed on **WCLIENT** (192.168.20.20). The agent transmits the following data to the server:

- Windows events
- System information
- Sysmon events
- FIM events
- Configuration data
- Inventory information

This communication allows Wazuh to centralize the endpoint's telemetry.

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17. Sysmon

Sysmon (System Monitor) was installed on WCLIENT. It enriches Windows events with additional information about system activity. Monitoring can notably cover:

- Process creation
- Network connections
- Image loading
- Modifications
- System information

A key event in the lab is **Event ID 1 — Process Creation**. Sysmon data is then visible in Wazuh.

```
PS C:\Users\Manager\Downloads\Sysmon> Get-WinEvent -FilterHashtable @{ LogName = "Microsoft-Windows-Sysmon
ational" ; Id = 1 } -MaxEvents 10


   ProviderName: Microsoft-Windows-Sysmon

TimeCreated                     Id LevelDisplayName Message
-----------                     -- ---------------- -------
9/18/2026 7:21:18 PM             1 Information      Process Create:...
9/18/2026 7:20:59 PM             1 Information      Process Create:...
9/18/2026 7:20:56 PM             1 Information      Process Create:...
9/18/2026 7:20:28 PM             1 Information      Process Create:...
9/18/2026 7:20:23 PM             1 Information      Process Create:...
9/18/2026 7:19:56 PM             1 Information      Process Create:...
9/18/2026 7:19:47 PM             1 Information      Process Create:...
9/18/2026 7:19:25 PM             1 Information      Process Create:...
9/18/2026 7:19:24 PM             1 Information      Process Create:...
9/18/2026 7:18:55 PM             1 Information      Process Create:...


PS C:\Users\Manager\Downloads\Sysmon>
```

Figure 13 — Sysmon Event ID 1 (Process Create) entries retrieved with Get-WinEvent

```
PS C:\WINDOWS\system32> Get-Service Sysmon64

Status   Name               DisplayName
------   ----               -----------
Running  Sysmon64           Sysmon64


PS C:\WINDOWS\system32> _
```

Figure 14 — Sysmon64 service in the running state

---

18. File Integrity Monitoring

File Integrity Monitoring (FIM) monitors changes made to files. A test directory was created for this purpose:

```
PS C:\WINDOWS\system32> New-Item -ItemType Directory -Path C:\SecureCorp-Test -Force


    Directory: C:\


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----         9/19/2026  10:57 PM                SecureCorp-Test


PS C:\WINDOWS\system32> _
```

Figure 15 — Creation of the C:\SecureCorp-Test test directory

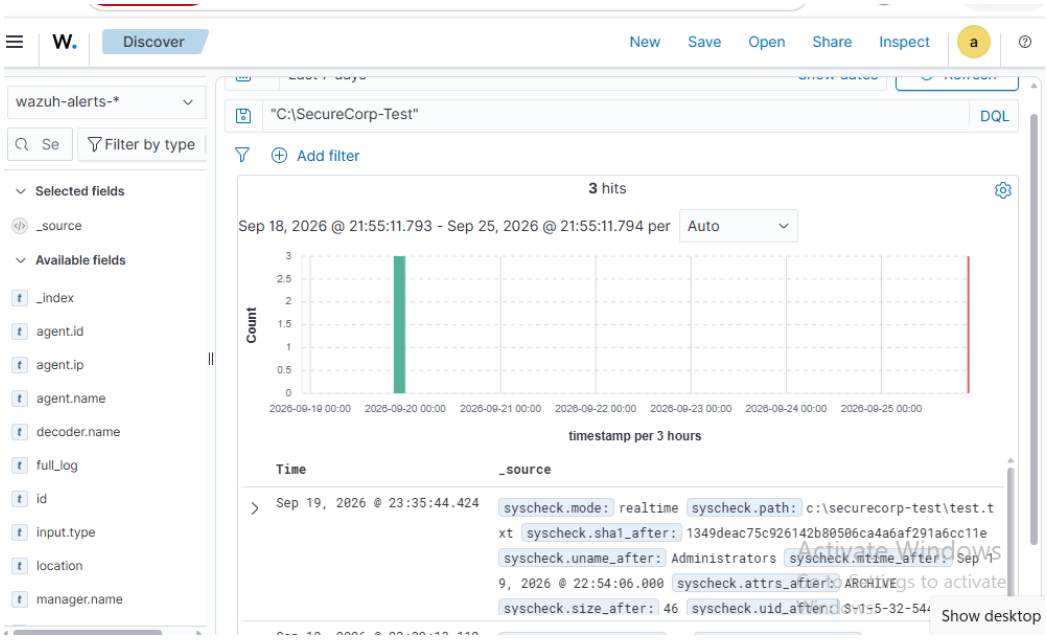


Figure 16 — FIM events for C:\SecureCorp-Test displayed in Wazuh Discover (3 hits)

This demonstrates that real-time monitoring of the directory is operational.

---

19. Security Configuration Assessment

The Wazuh SCA module assesses the security configuration of the system. In the lab, the following benchmark is available:

CIS Microsoft Windows 11 Enterprise Benchmark

This feature provides a structured view of selected security controls on the workstation. SCA notably allows:

- Identifying compliant controls
- Identifying non-compliant controls
- Analyzing the configuration
- Tracking improvements in security posture

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20. Syscollector

Syscollector provides inventory information about the system. The configuration used notably collects:

- Hardware
- Operating system
- Network
- Packages
- Ports
- Processes
- Users

- Groups
- Services
- Browser extensions

The scan interval is approximately one hour, with an additional scan at startup. This feature provides a view of the state of the monitored workstation.

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21. Windows Security Events

Collecting Windows events is an essential part of the lab. The following events were studied.

21.1 Event 4625 — Failed Logon

Event ID 4625 corresponds to a failed authentication attempt. In Wazuh, the observed data notably includes the following fields:

```
data.win.system.channel
data.win.system.computer
data.win.system.eventID
data.win.system.eventRecordID
data.win.system.workstationName
```

The system has therefore demonstrated its ability to collect Windows authentication events.

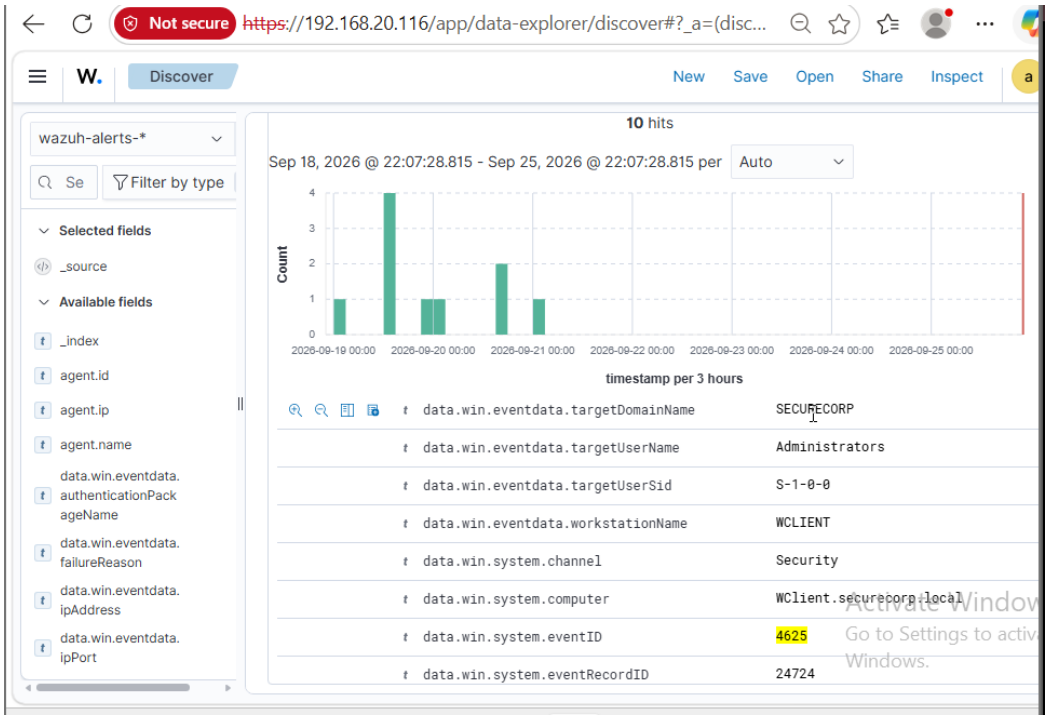


Figure 17 — Event 4625 in Wazuh Discover (10 hits over the last seven days)

21.2 Event 4624 — Successful Logon

Event ID 4624 corresponds to a successful authentication and provides additional visibility into logon sessions.

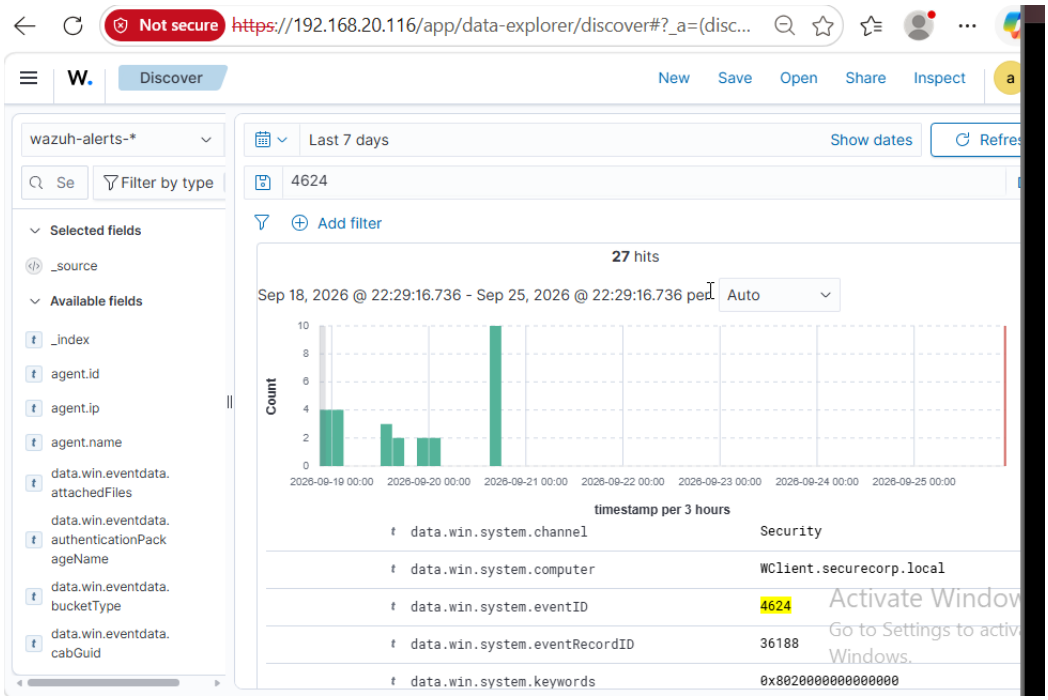


Figure 18 — Event 4624 in Wazuh Discover (27 hits over the last seven days)

21.3 Event 4688 — Process Creation

Event ID 4688 corresponds to the creation of a Windows process. It is particularly useful for monitoring activity executed on the endpoint.

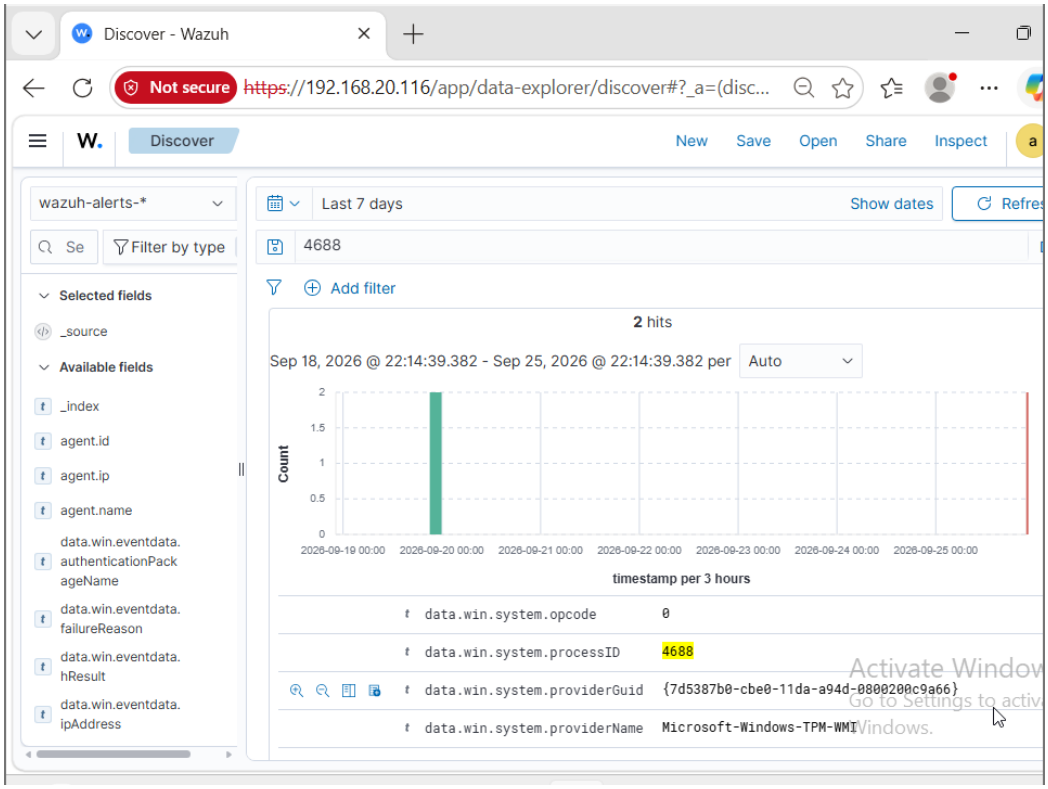


Figure 19 — Event 4688 in Wazuh Discover (2 hits over the last seven days)

21.4 Event 4720 — User Account Created

Event ID 4720 is generated when a user account is created. Within the lab, a test account can be created to generate this event.

21.5 Event 4728 — Member Added to a Security-Enabled Global Group

This event can be used to observe changes made to Active Directory security groups.

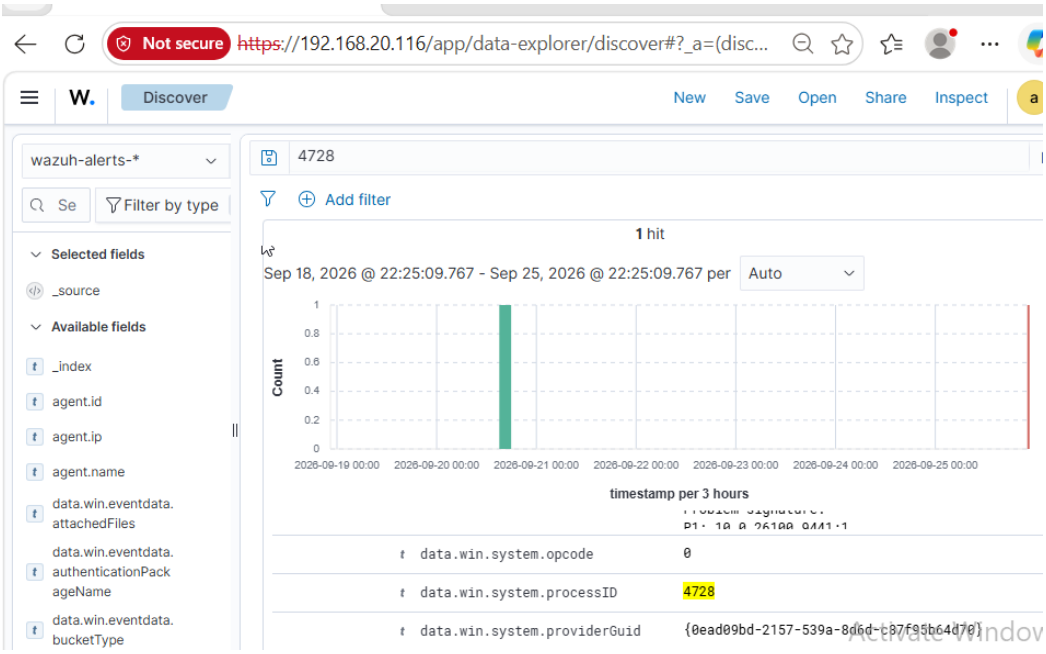


Figure 20 — Event 4728 in Wazuh Discover (1 hit)

21.6 Event 4732 — Member Added to a Security-Enabled Local Group

This event allows monitoring of changes made to local security groups.

---

22. PowerShell Monitoring

PowerShell is an important source of Windows telemetry. The log used is:

```
Microsoft-Windows-PowerShell/Operational
```

Legitimate commands were run to generate and verify the events:

- Get-Process
- Get-Service

The resulting PowerShell events were then searched for in Wazuh. This collection demonstrates the ability to monitor PowerShell activity.

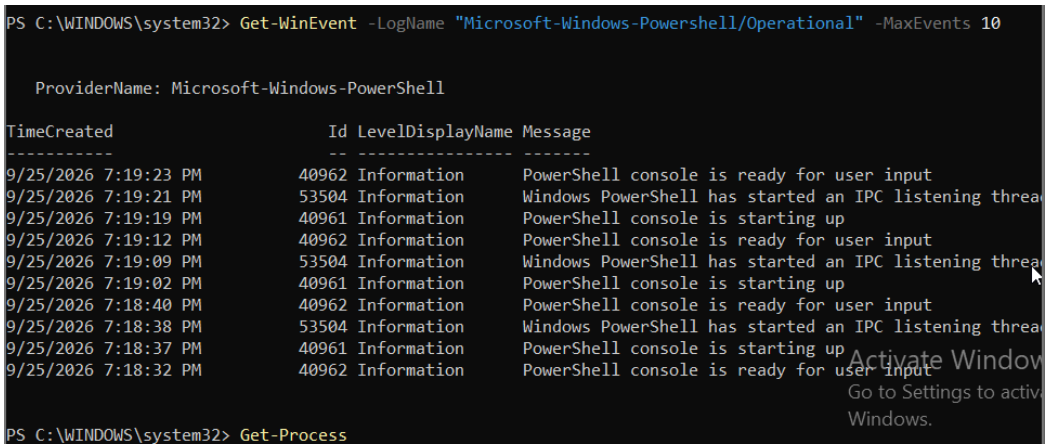


Figure 21 — PowerShell Operational log entries retrieved with Get-WinEvent

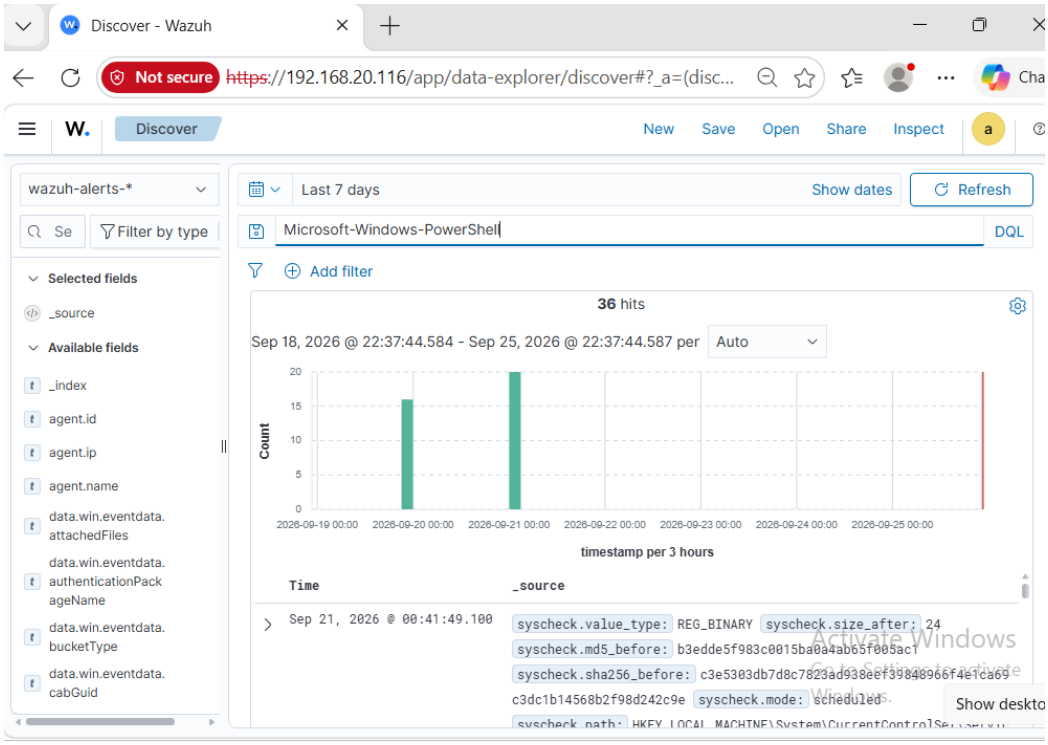


Figure 22 — PowerShell events in Wazuh Discover (36 hits over the last seven days)

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23. Network Reconnaissance Scenario

23.1 Objective

The first security scenario consists of performing reconnaissance against WCLIENT from Kali.

```
Kali (192.168.10.100)
   |  Nmap
   ▼
OPNsense
   |
   ▼
WCLIENT (192.168.20.20)
```

23.2 Service Scan

The command used is:

```
nmap -Pn -sV -sC 192.168.20.20
```

This command identifies accessible ports and attempts to determine the services running on them and their versions.

Port	State	Service	Version
135/tcp	open	msrpc	Microsoft Windows RPC
139/tcp	open	netbios-ssn	Microsoft Windows netbios-ssn
445/tcp	open	microsoft-ds	—

The remaining 997 scanned ports were filtered. The SMB2 security mode reported by the scan indicates that message signing is enabled and required.

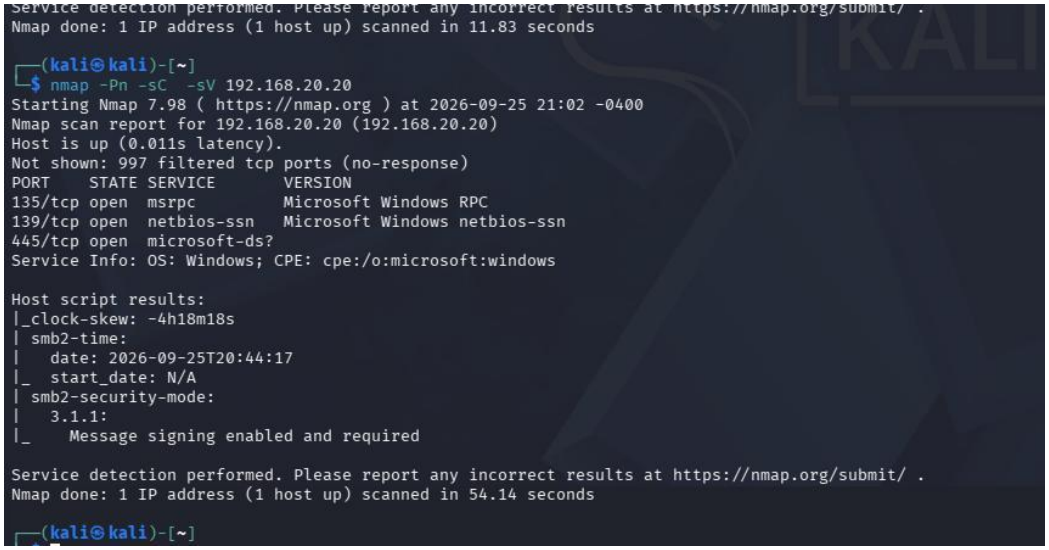


Figure 23 — Nmap service scan of WCLIENT from Kali

23.3 Analysis

Reconnaissance identifies the network surface visible from the analysis network. This phase is considered a preliminary step to the security analysis. The project does not, however, aim to exploit the identified services.

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24. MITRE ATT&CK Mapping

The lab's activities can be mapped to specific MITRE ATT&CK techniques. This is a documentary mapping and does not imply that every technique was executed in full.

Activity	Technique	Description
Network scanning	T1046	Network Service Scanning
PowerShell	T1059.001	Command and Scripting Interpreter: PowerShell

Activity	Technique	Description
Process discovery	T1057	Process Discovery
Account discovery	T1087	Account Discovery
Use of a valid account	T1078	Valid Accounts (depending on the scenario)
Windows logon	Events 4624 / 4625	Authentication telemetry
File monitoring	Wazuh FIM	Integrity monitoring

---

25. Results Obtained

The project made it possible to put several security capabilities in place.

Network

- Segmentation of the virtual networks
- Routing with OPNsense
- Separation between the attack network and the corporate network
- Controlled communication between the networks

Active Directory

- Domain creation
- Organization of users and groups
- Organizational units
- Group Policy Objects
- Analysis with BloodHound

Endpoint

- Installation of the Wazuh agent
- Installation of Sysmon
- Collection of Windows events
- FIM monitoring
- Syscollector inventory
- SCA assessment

Detection

The following Windows events were studied: 4624, 4625, 4688, 4720, 4728 and 4732, together with PowerShell, Sysmon, FIM and SCA data.

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26. Security Analysis

The lab shows that a monitoring solution should not rely on a single source of data. Fuller visibility is obtained by combining:

- Windows events
- Sysmon
- File Integrity Monitoring
- Security Configuration Assessment
- Syscollector
- Network telemetry

This approach provides complementary perspectives on the activity of an endpoint:

Activity	Visible through
Nmap scan	Network activity
PowerShell execution	Process and command activity
File modification	File Integrity Monitoring
Authentication failure	Windows Security event
System configuration	Security Configuration Assessment

Correlating this information is an important foundation for SOC analysis.

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27. Conclusion

The SecureCorp Cybersecurity Lab project made it possible to build a virtualized cybersecurity environment that reproduces several components of an enterprise infrastructure. The lab combines network security, Active Directory, endpoint security, SIEM and offensive security.

The integration of OPNsense makes it possible to segment and route the different virtual networks. Active Directory provides a centralized infrastructure for managing users, groups and computers, and BloodHound makes it possible to analyze the relationships between Active Directory objects.

Wazuh is the core of the security monitoring. It collects information from WCLIENT and makes it possible to analyze Windows, Sysmon, FIM, SCA and Syscollector events. The reconnaissance and controlled authentication scenarios demonstrate how activity generated in the lab can be observed at both the endpoint and the monitoring-platform level.

The project thus provides a practical foundation for studying how a simplified SOC environment operates:

```
Reconnaissance
   ↓
Activity
   ↓
Endpoint
   ↓
Telemetry
   ↓
Wazuh
   ↓
Detection
   ↓
Analysis
```

The main objective of the lab is therefore not only to carry out attacks, but above all to understand how security activity is generated, collected, analyzed and documented.